

WEEK2-PARTICLE-SWARM-OPTIMIZATION

```
import random

particles = [10, 15, 30, 40, 20]
velocity = [0] * 5
pbest = particles.copy()

target = 25
gbest = min(pbest, key=lambda x: abs(x - target))

for i in range(10):
    for j in range(5):

        r1 = random.random()
        r2 = random.random()

        velocity[j] = (0.7 * velocity[j] +
                        1.5 * r1 * (pbest[j] - particles[j]) +
                        1.5 * r2 * (gbest - particles[j]))

        particles[j] += velocity[j]

        if abs(particles[j] - target) < abs(pbest[j] - target):
            pbest[j] = particles[j]

        if abs(pbest[j] - target) < abs(gbest - target):
            gbest = pbest[j]

    print("Iteration", i + 1, "Best Temperature:", round(gbest, 2))

print("\nTarget Temperature:", target)
print("Optimal Temperature:", round(gbest, 2))
```

OUTPUT

```
*** Iteration 1 Best Temperature: 26.16
    Iteration 2 Best Temperature: 26.16
    Iteration 3 Best Temperature: 24.82
    Iteration 4 Best Temperature: 24.82
    Iteration 5 Best Temperature: 25.1
    Iteration 6 Best Temperature: 25.1
    Iteration 7 Best Temperature: 25.1
    Iteration 8 Best Temperature: 25.1
    Iteration 9 Best Temperature: 25.1
    Iteration 10 Best Temperature: 25.1

    Target Temperature: 25
    Optimal Temperature: 25.1
```